

Retail Marketing Campaign Analytics

A retail company wants to analyze customer buying behavior and marketing campaign performance using Snowflake sample data.

The company has three teams:

1. Marketing Analysts
2. Finance Team
3. ETL Developers

Each team should use a separate warehouse. The company also wants to improve query performance using:

1. Warehouse scaling
2. Query result cache
3. Cluster key on customer sales date

Use Snowflake sample data from:

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM

Questions

Part 1: Warehouse Setup

Create three warehouses:

1. MARKETING_WH for marketing analysts
2. FINANCE_WH for finance team
3. ETL_WH for ETL developers

Use different warehouse sizes based on workload.

```
1 CREATE OR REPLACE WAREHOUSE MARKETING_WH
2 WAREHOUSE_SIZE = 'XSMALL'
3 AUTO_SUSPEND = 60
4 AUTO_RESUME = TRUE;
5
6 CREATE OR REPLACE WAREHOUSE FINANCE_WH
7 WAREHOUSE_SIZE = 'SMALL'
8 AUTO_SUSPEND = 60
9 AUTO_RESUME = TRUE;
10
11
12 CREATE OR REPLACE WAREHOUSE ETL_WH
13 WAREHOUSE_SIZE = 'MEDIUM'
14 AUTO_SUSPEND = 60
15 AUTO_RESUME = TRUE;
```

Results (1 minute ago)

status
Warehouse ETL_WH successfully created.

ETL_WH	Standard	Medium
FINANCE_WH	Standard	Small
MARKETING_WH	Standard	X-Small

Part 2: Role Setup

Create three roles:

- MARKETING_ANALYST
- FINANCE_USER
- ETL_DEVELOPER

Grant each role access only to its assigned warehouse.

```
16
17 CREATE OR REPLACE ROLE MARKETING_ANALYST;
18 CREATE OR REPLACE ROLE FINANCE_USER;
19 CREATE OR REPLACE ROLE ETL_DEVELOPER;
```

Results (just now)

status
Role ETL_DEVELOPER successfully created.

```

16
17     CREATE OR REPLACE ROLE MARKETING_ANALYST;
18
19     CREATE OR REPLACE ROLE FINANCE_USER;
20
21     CREATE OR REPLACE ROLE ETL_DEVELOPER;
22
23     GRANT USAGE ON WAREHOUSE MARKETING_WH TO ROLE MARKETING_ANALYST;
24
25     GRANT USAGE ON WAREHOUSE FINANCE_WH TO ROLE FINANCE_USER;
26
27     GRANT USAGE ON WAREHOUSE ETL_WH TO ROLE ETL_DEVELOPER;
28

```

Part 3: User Setup

Create users:

1. MKT_USER1
2. FIN_USER1
3. ETL_USER1

Set default role and default warehouse for each user.

```

28
29     CREATE OR REPLACE USER MKT_USER1
30     PASSWORD = 'Password@123'
31     DEFAULT_ROLE = MARKETING_ANALYST
32     DEFAULT_WAREHOUSE = MARKETING_WH
33     MUST_CHANGE_PASSWORD = FALSE;
34
35     CREATE OR REPLACE USER FIN_USER1
36     PASSWORD = 'Password@123'
37     DEFAULT_ROLE = FINANCE_USER
38     DEFAULT_WAREHOUSE = FINANCE_WH
39     MUST_CHANGE_PASSWORD = FALSE;
40
41     CREATE OR REPLACE USER ETL_USER1
42     PASSWORD = 'Password@123'
43     DEFAULT_ROLE = ETL_DEVELOPER
44     ✨ DEFAULT_WAREHOUSE = ETL_WH
45     MUST_CHANGE_PASSWORD = FALSE;
46

```

```
46
47     GRANT ROLE MARKETING_ANALYST TO USER MKT_USER1;
48
49     GRANT ROLE FINANCE_USER TO USER FIN_USER1;
50
51     GRANT ROLE ETL_DEVELOPER TO USER ETL_USER1;
52
53     SHOW GRANTS TO USER MKT_USER1;
54
55     SHOW GRANTS TO USER FIN_USER1;
56
57     SHOW GRANTS TO USER ETL_USER1;
58
59     SHOW USERS;
60
```

Results (just now)

Table Chart Pivot

1 row 31ms

	created_on	privilege	granted_on	name	role	granted_to	grantee_name
1	2026-05-15 22:47:15	USAGE	ROLE	ETL_DEVELOPER	ETL_DEVELOPER	USER	ETL_USER1

Part 4: Create Performance Demo Database

Create a database called:

RETAIL_MARKETING_DB

Create a table called:

CAMPAIGN_SALES

using Snowflake sample tables.

My Workspace > assignment.sql

ACCOUNTADMIN • ETL_WH (Medium) RETAIL_MARKETING_DB PUBLIC Share ...

```
56
57     SHOW GRANTS TO USER ETL_USER1;
58
59     SHOW USERS;
60
61     CREATE OR REPLACE DATABASE RETAIL_MARKETING_DB;
62     USE DATABASE RETAIL_MARKETING_DB;
63     CREATE OR REPLACE SCHEMA PUBLIC;
64
65
66     Ctrl+I to generate
```

Results (just now)

Table Chart Pivot

1 row 69ms

	status
1	Schema PUBLIC successfully created.

My Workspace > assignment.sql

ACCOUNTADMIN ETL_WH (Medium) RETAIL_MARKETING_DB PUBLIC Share ...

```

65
66 CREATE OR REPLACE TABLE CAMPAIGN_SALES AS
67 SELECT
68     C.C_CUSTOMER_SK,
69     C.C_FIRST_NAME,
70     S.SS_SOLD_DATE_SK,
71     D.D_DATE AS SALE_DATE,
72     I.I_ITEM_DESC,
73     S.SS_NET_PAID AS SALES_AMOUNT
74 FROM SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES S
75 JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER C
76     ON S.SS_CUSTOMER_SK = C.C_CUSTOMER_SK
77 JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM D
78     ON S.SS_SOLD_DATE_SK = D.D_DATE_SK
79 JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM I
80     ON S.SS_ITEM_SK = I.I_ITEM_SK
81 LIMIT 100000000;
82
83 | Ctrl+I to generate

```

Results (just now)

Table Chart Pivot

1 row 9.4s

#	status
1	Table CAMPAIGN_SALES successfully created.

```

CREATE OR REPLACE TABLE CAMPAIGN_SALES AS
SELECT
    C.C_CUSTOMER_SK,
    C.C_FIRST_NAME,
    S.SS_SOLD_DATE_SK,
    D.D_DATE AS SALE_DATE,
    I.I_ITEM_DESC,
    S.SS_NET_PAID AS SALES_AMOUNT
FROM SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES S
JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER C
    ON S.SS_CUSTOMER_SK = C.C_CUSTOMER_SK
JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM D
    ON S.SS_SOLD_DATE_SK = D.D_DATE_SK
JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM I
    ON S.SS_ITEM_SK = I.I_ITEM_SK
LIMIT 100000000;

```

SELECT * FROM CAMPAIGN_SALES LIMIT 10;

Add to Chat Explain Quick edit Format

Results (just now)

Table Chart Pivot

10 rows 715ms

# C_CUSTOMER_SK	C_FIRST_NAME	SS_SOLD_DATE_SK	SALE_DATE	I_ITEM_DESC	SALES_AMOUNT
38037792	Roland	2451157	12/9/1998	Advantages should echo besides to the qualities. Imp...	12.21
56497925	Brian	2451157	12/9/1998	Alleged plants ought to make	2401.98
56497925	Brian	2451157	1998-12-09	+8 more	
38037792	Roland	2451157	1998-12-09	Fat trees used to see also new components. Children shall not ch	1523.80
38037792	Roland	2451157	1998-12-09	Mild, long weeks disappoint plants. Official eyes suggest with a d	1064.48
38037792	Roland	2451157	1998-12-09	As existing teachers open already awkward, main reasons. New v	12.21
38037792	Roland	2451157	1998-12-09	Important, thin losses drop highly long resources. Various bills giv	261.45

Part 5: Grant Permissions

Grant proper database, schema, and table permissions to each role.

Marketing and Finance can only read data.

ETL Developer can create and modify tables.

```

GRANT USAGE ON DATABASE RETAIL_MARKETING_DB TO ROLE MARKETING_ANALYST;

GRANT USAGE ON DATABASE RETAIL_MARKETING_DB TO ROLE FINANCE_USER;

GRANT USAGE ON DATABASE RETAIL_MARKETING_DB TO ROLE ETL_DEVELOPER;

GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC TO ROLE MARKETING_ANALYST;

GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC TO ROLE FINANCE_USER;

GRANT USAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC TO ROLE ETL_DEVELOPER;

GRANT SELECT ON TABLE RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES TO ROLE MARKETING_ANALYST;

GRANT SELECT ON TABLE RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES TO ROLE FINANCE_USER;

GRANT ALL PRIVILEGES ON TABLE RETAIL_MARKETING_DB.PUBLIC.CAMPAIGN_SALES TO ROLE ETL_DEVELOPER;

GRANT CREATE TABLE, CREATE VIEW, CREATE STAGE ON SCHEMA RETAIL_MARKETING_DB.PUBLIC
TO ROLE ETL_DEVELOPER;

```

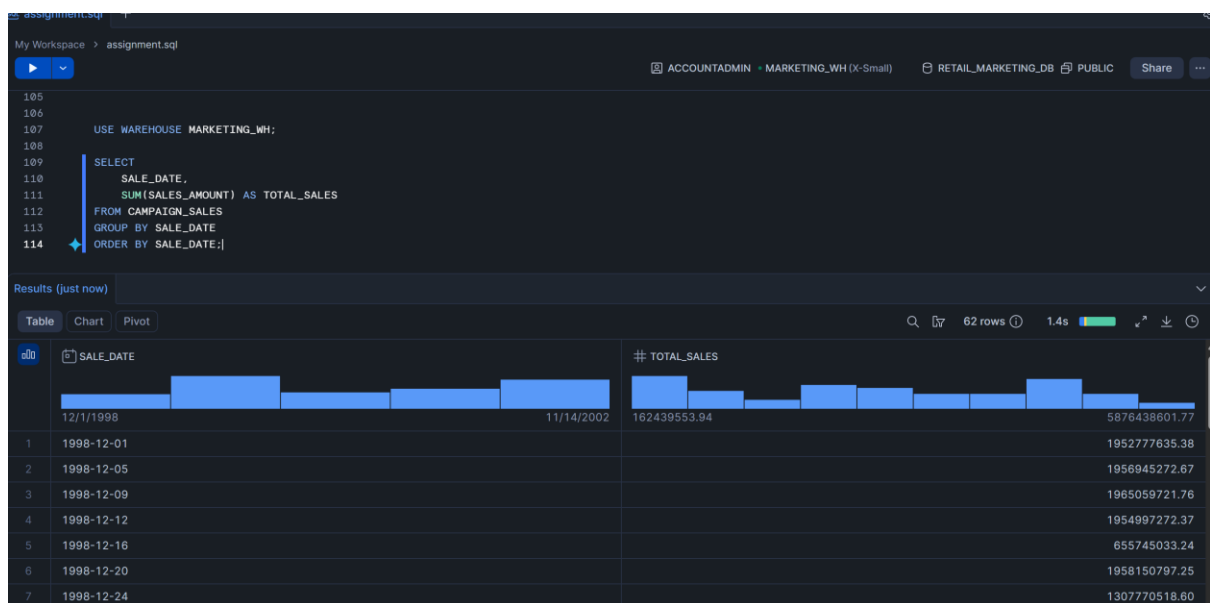
Part 6: Performance Testing

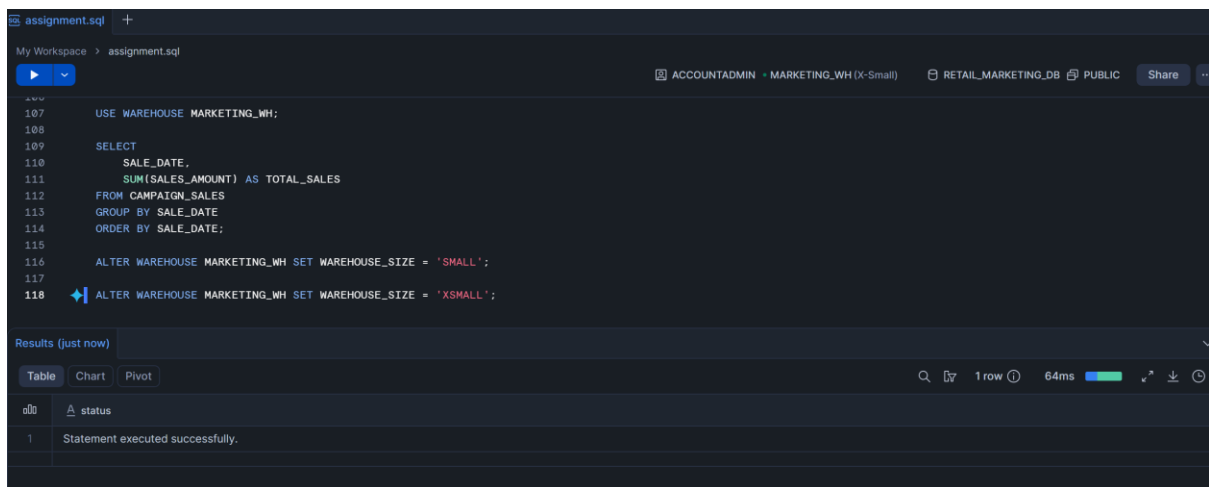
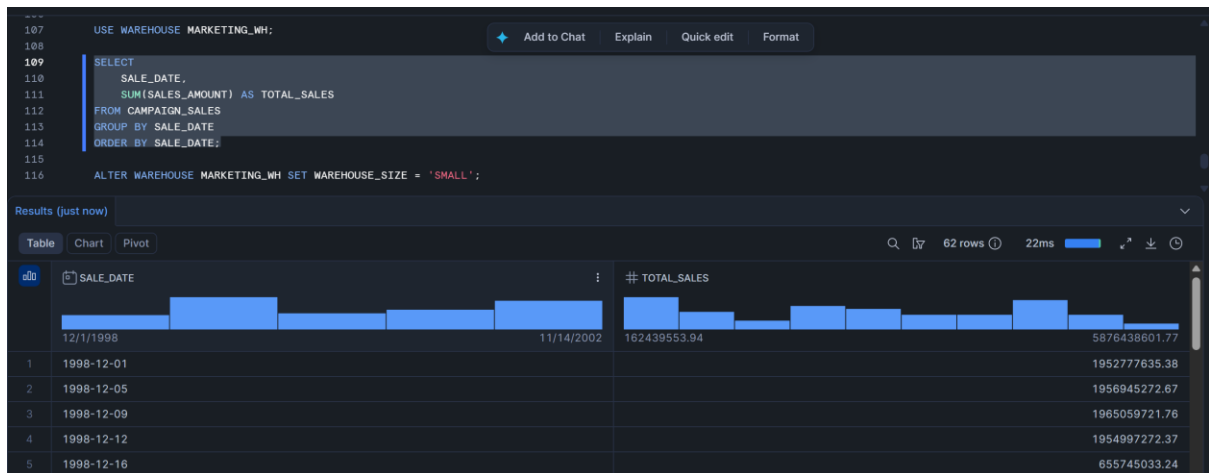
Run a sales analysis query using XSMALL warehouse.

Then scale up the warehouse to SMALL.

Compare performance.

Then scale it down again.

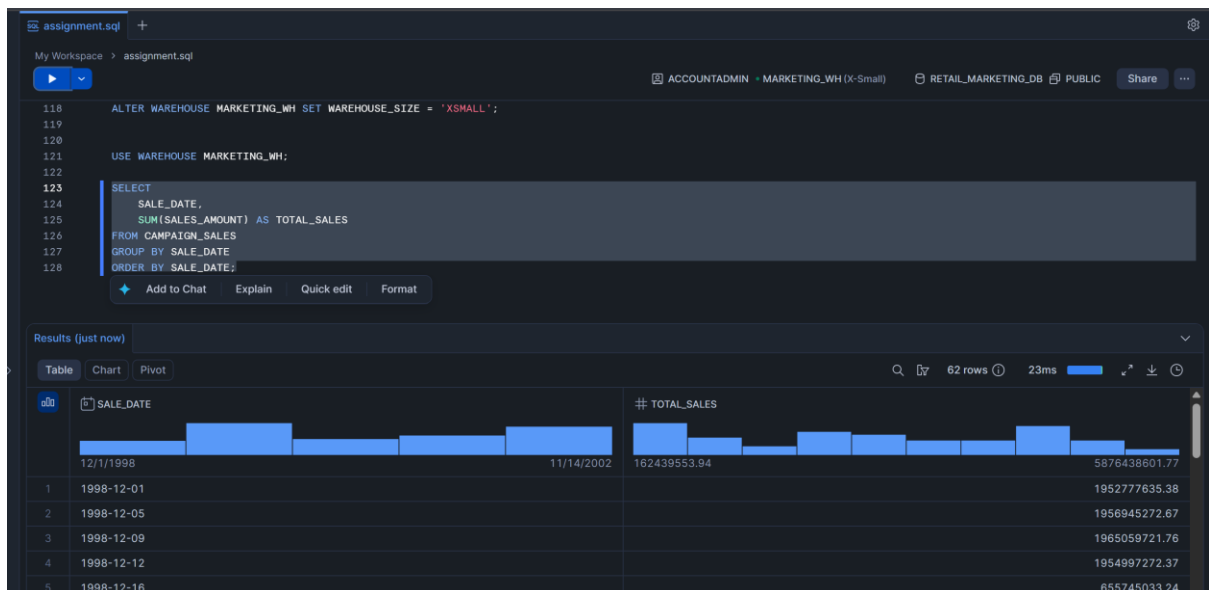


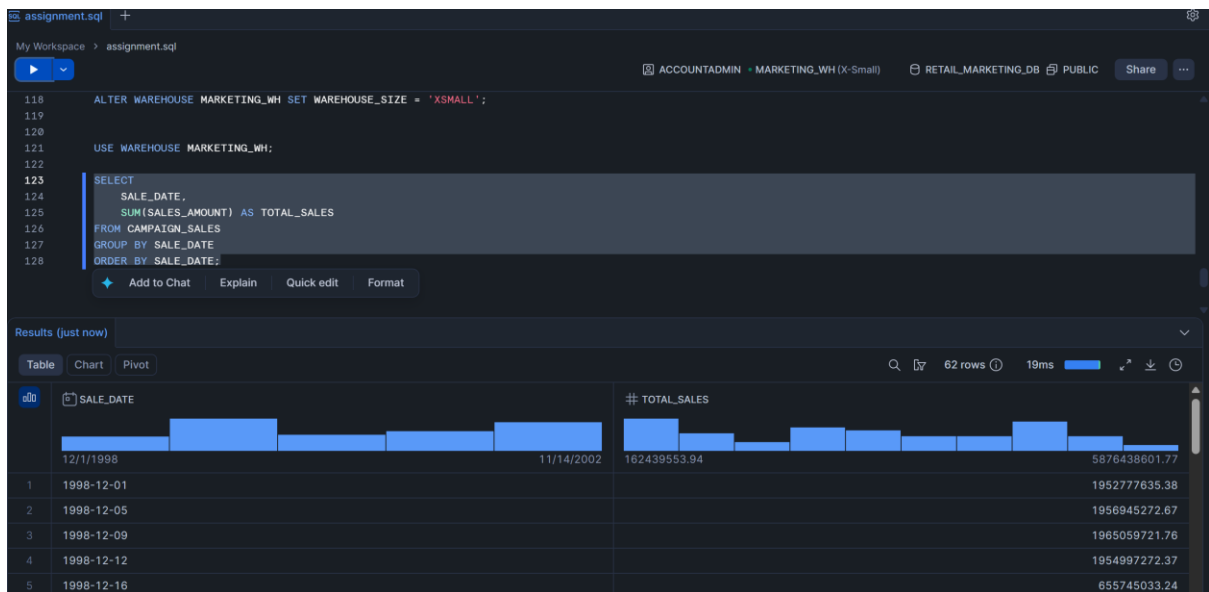


Part 7: Cache Testing

Run the same query twice.

Observe why the second query is faster.





Part 8: Cluster Key Testing

Run a date-filter query before clustering.

Add cluster key on SALE_DATE.

Run the same query again.

Compare performance.

